

# GROCERY BILLING RECEIPT GENERATOR

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*(Using Core Java & OOP Concepts)*

Under the Guidance of

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**Object Oriented Programming With Java**

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## ABSTRACT

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This project focuses on designing and implementing a Grocery Billing Receipt Generator using Object-Oriented Programming (OOP) concepts in Java. The system is developed to automate the billing process in retail grocery stores, reducing manual errors and improving efficiency.

The application allows users to enter product details such as item name, price, and quantity. It then calculates the total cost, applies tax, and generates a well-structured receipt.

This project demonstrates real-world use of OOP concepts such as classes, objects, and aggregation. It also uses efficient string handling techniques like `StringBuilder` to generate formatted output.

# INTRODUCTION

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Billing systems are an essential part of retail businesses. Manual billing is time-consuming and error-prone. This project provides a simple automated solution using Java.

The Grocery Billing Receipt Generator helps in managing item details, calculating totals automatically, applying tax, and generating formatted receipts.

It uses Object-Oriented Programming concepts like classes, objects, and aggregation where the Bill class contains multiple Item objects.

# OBJECTIVES

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- Store product details (name, price, quantity)
- Calculate total bill automatically
- Apply tax (GST)
- Generate clean receipt output
- Implement OOP concepts

### 3. SCOPE

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This project is suitable for small grocery shops and basic retail systems. It can be extended in the future by integrating databases, graphical user interfaces, and advanced billing features.

### 4. SYSTEM REQUIREMENTS

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Hardware: Computer with **4GB** RAM

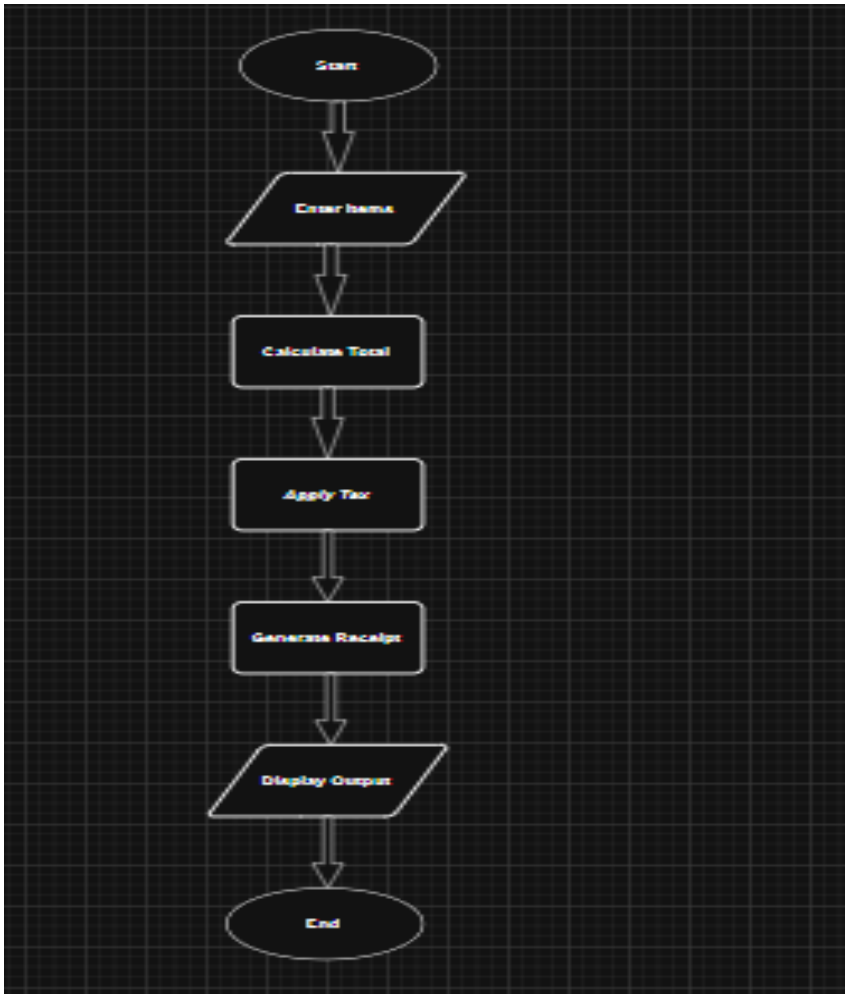
Software: **Java JDK, VS Code / Eclipse**

### 5. METHODOLOGY

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1. User enters item details
2. Items are stored using Item class
3. Items are passed to Bill class
4. Total is calculated
5. Tax is applied
6. Receipt is generated
7. Output is displayed

# FLOWCHART



| Step            | Shape         | Purpose                                                                                           |
|-----------------|---------------|---------------------------------------------------------------------------------------------------|
| Start           | Oval          | The entry point of the program.                                                                   |
| Enter Items     | Parallelogram | <b>Input:</b> Taking data from the user or a scanner.                                             |
| Calculate Total | Rectangle     | <b>Process:</b> Summing up the prices ( $\text{\$Price\_1} + \text{Price\_2} = \text{Total\$}$ ). |
| Apply Tax       | Rectangle     | <b>Process:</b> Calculating tax using a formula, e.g., $\text{\$Total} \times 1.08\$$ .           |

|                  |               |                                                     |
|------------------|---------------|-----------------------------------------------------|
| Generate Receipt | Rectangle     | Process: Formatting the data into a bill structure. |
| Display Output   | Parallelogram | Output: Printing the receipt to a screen or paper.  |
| End              | Oval          | The exit point of the program.                      |

## FUTURE ENHANCEMENTS

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This project can be improved by adding advanced features such as a graphical user interface using Java Swing, database integration for storing records, barcode scanning, and discount calculation. It can also be converted into a full retail management system.

## GANTT CHART

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| Task     | Duration |
|----------|----------|
| Planning | 1 Day    |
| Design   | 2 Days   |
| Coding   | 1 Day    |
| Testing  | 1 Day    |

## CONCLUSION

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*The project successfully demonstrates the implementation of Object-Oriented Programming concepts in Java. It provides an efficient and user-friendly way to generate grocery bills. The system reduces manual effort and ensures accurate calculations. It can be further enhanced with additional features to make it more powerful and scalable.*

## REFERENCES

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